

FOCUS

Quantum Computing

Quantum Machine Learning: An Overview



Quantum machine learning (QML) is a burgeoning field at the intersection of quantum computing and artificial intelligence. In recent years, the

integration of quantum mechanics with machine learning algorithms has sparked substantial interest among researchers and technologists alike. Here's a quick look at the essentials of creating quantum algorithms for AI models, their practical use cases on open source platforms, and best practices for implementing these advanced algorithms... [Read more on page.....34](#)

The Role of Open Source in Accelerating Quantum AI



Here's an overview of how open source frameworks are being utilised to build quantum machine learning models, including quantum neural networks and quantum kernel

methods. The challenges and future directions in the quantum AI landscape are also discussed. [Read more on page.....38](#)

From Bits to Qubits: The Growth Story of Quantum Computing



Quantum computing may still be in the early stages of evolution, but its potential impact on everyday life is significant. We delve into the key concepts behind it, the

reasons for its rapid growth, and how global advancements are shaping its future. [Read more on page.....49](#)

Cirq: The Open Source Framework for Programming Quantum Computers

Explore the key features, capabilities, and impact of Cirq, an open source quantum computing framework developed by Google, on the quantum programming landscape.

[Read more on page.....42](#)

How Quantum Computing Differs from Classical Computing

Despite being in its infancy, quantum computing has numerous potential applications in modelling, cybersecurity, AI/ML, and other fields. But how do quantum and classical computing compare with each other? Let's find out... [Read more on page.....54](#)

Quantum Computing: Harnessing Open Source for Innovation and Accessibility

We explore how open source initiatives are shaping the future of quantum computing, making it more accessible and driving innovation through collaboration.

[Read more on page.....57](#)

Popular Open Source Toolkits for Quantum Machine Learning

Quantum machine learning is becoming increasingly popular due to its ability to solve the complex problems of the AI age. Here are a few open source libraries and frameworks that help with quantum computations.

[Read more on page.....61](#)